SIMATS ENGINEERING
Department of Electrical and Electronics Engineering
ASSESSMENT TOOL – 2

Engineer to Excel

Course Code:	ECA1418	Course Title:	Embedded Systems
CO Assessed:	CO : 1	Assessment:	At-2
Weightage:		Total Marks:	
CO1 Statement:			
Student Name:	M. Nandhini	Reg. No.:	192472318
Section / Batch:	3rd Year (Slot-B)	Date:	24-07-26

Question Type	Description	Questions	Total Marks
	Design Task	Q – Q	100

Code of Conduct

I M. Nandhini (192472318) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M. Nandhini

RUBRICS FOR MARKS CALCULATION

Task	(%)	(%)	(%)	(%)	Total Marks (Each -)
					20
				Overall Total ( )	

Faculty In-charge	Course Coordinator	Head of Department
Name: Signature:	Signature:	Signature:
Date: 24/7/26	Date:	Date:

Assessment Tool - 2

Name - M. Nandhini

Reg. No - 192472318

Course Name - Embedded Systems

Course Code - ECA1418

Faculty Name - Dr. C M ARUN KUMAR.

Performance Analysis and Redesign of a CAN-Based Smart Home Automation System

Introduction:

A smart Home Automation System uses the controller Area Network (CAN) protocol to connect devices such as lighting, HVAC, security sensors, door locks, and user interfaces. The system contains 35 nodes communicating at 250 kbps. Recently users reported a 3-second delay when controlling home devices. The task is to analyze the problem, identify the bottleneck, and redesign the message scheduling strategy to reduce the response time to less than 200 ms without replacing any hardware.

Understanding of the Existing Design:

The smart home automation system consists of different embedded nodes connected through a CAN bus. Each node performs a specific task and periodically sends and periodically sends data over the communication network.

Node Configuration:

Node Type	No. of Nodes	Function	Message Period
Lighting Controllers	15	Controls lights	100 ms
Security Sensors	10	motion and intrusion detection	50 ms
Door Locks	5	Lock and unlock doors	Event-based
HVAC controller	1	Temperature control	5 ms
User interfaces	2	Mobile application	200 ms
Gateway	2	Internet Communication	100 ms

Evaluation of the Existing Design:

Strengths: The current design has several advantages:

- CAN protocol provides reliable communication.
- Built-in-error detection and error recovery improve System reliability.
- Multiple devices can share the same communication bus.
- Priority-based arbitration allows important messages to be transmited first.

Weakneses: The HVAC controller transmits data every 5 milliseconds which means it sends 200 messages every second. Because these messages have high priority, they frequently win CAN arbitration.

As a result:

- Lower-Priority nodes wait for transmission.
- Communication queues become longer.
- Bus utilization increases significantly.
- Other Commands experience large delays.

Identified performance Bottleneck:

The HVAC controller is identified as the main Source of network Congestion because of its high-frequency transmission.

Bus load Analysis:

CAN network work efficiently when the bus load remains below 70%. In this system, the HVAC controller sends 200 messages per second, increasing the bus utilization to approximately 90-95%. As the bus becomes overloaded.

- Arbitration delays increase.
- Message queues becomes longer.
- Low-priority devices wait for transmission.

Node Type	Traffic Contribution	Bus Load
Lighting	Low	Low
Security Sensors	Medium	Moderate
Door Locks	Very Low	Low
HVAC Controller	Very High	Highest
User Interface	Low	Low
Gateway	Medium	Moderate.

The HVAC controller occupies a large percentage of the CAN bandwidth due to its frequent transmissions. Even though there is only one HVAC node, it generates much more traffic than many other nodes combined.

Latency Escalation Through CAN Arbitration:

The CAN protocol uses arbitration to determine which node can transmit first.

Communication flow

HVAC Node (5ms)
↓
200 messages Every Second
↓
High CAN Priority
↓
Wins Bus Arbitration
↓
Other Nodes Wait
↓
Message Queue Increases
↓
High Bus Utilization
↓
Response Delay Reaches 3 seconds.

Expected Performance After Redesign:

Parameter	Existing System	Redesigned System
HVAC Message Period	5ms	50ms
HVAC Messages/second	200	20
Bus Utilization	90-95%	55-60%
Arbitration Delay	High	Low
User Response Time	3 seconds	Below 200ms
Hardware Cost	No change	No change

The redesigned communication schedule reduces unnecessary traffic while maintaining the existing CAN hardware.

Conclusion:

The analysis clearly identifies the HVAC controller as the major performance bottleneck in the CAN-based Smart Home Automation System. Its 5ms transmission period generates excessive network traffic, resulting in high bus utilization, frequent arbitration delays, and long message queues. Consequently, homeowners experience response delays of approximately 3 seconds.

The HVAC controller is the main cause of network congestion because it sends messages too frequently. This increases CAN bus utilization and delays other messages. By increasing the HVAC transmission interval, using event-driven communication, and optimizing message priorities, the response time can be reduced from 3 sec to less than 200 ms without replacing any hardware.